Singapore Criminal Law Advisory System

Muhammad Asyraf Bin Omar*, **Tham Hao Wei***, **Umali Mike Guil Anonuevo***, **Wang Jingxi Jeslyn***
Singapore University of Technology and Design
{1006938, 1006936, 1006641, 1007067}@mymail.sutd.edu.sg

Abstract

This paper presents the design, implementation, and empirical evaluation of a large language model (LLM)-based legal advisory system for Singapore criminal law. The system enables users to retrieve and synthesise relevant criminal law principles, statutory provisions, and case precedents from Singapore Supreme Court judgments into structured advisory responses, while operating entirely under local deployment constraints. We experimented with three approaches — a multi-agent retrieval-augmented generation (RAG) pipeline, parameter-efficient fine-tuning via QLoRA on Qwen2.5-1.5B and 3B models, and a commercial API baseline (Claude) as an upper-bound reference — before selecting multi-agent RAG as the final architecture. The system is supported by a domain-specific corpus of 876 Singapore Supreme Court criminal judgments, yielding 83,322 indexed text chunks. Corpus expansion from 500 to 876 judgments drove the most significant performance gains, improving retrieval hit rate from 0.750 to 0.950 and mean cosine similarity from 0.583 to 0.664, demonstrating that domain corpus coverage is the primary driver of retrieval quality. These findings offer practical insights into building domain-specific legal AI systems under open-source, locally controlled deployment conditions.

1 Introduction

Large language models have significant potential in legal applications (Lai et al., 2024), but their direct use in high-stakes domains remains limited by hallucination, weak source grounding, and insufficient domain specificity. These limitations are especially problematic in Singapore criminal law, where users need answers that are tied closely to statutory provisions, procedural rules, sentencing principles, and case precedent. This project addresses that gap by developing a domain-specific Singapore Criminal Law Advisory System that provides structured, source-grounded guidance on what legal issues, procedures, and likely outcomes may arise in a criminal law scenario.

The system is not intended to replace a lawyer or function as a definitive legal authority. Instead, it is designed as an advisory and decision-support tool that helps a user and their counsel quickly understand what is likely to happen in a given criminal matter, including:

- what offences or legal issues may be involved,
- what procedural steps may follow,
- what sentencing considerations may arise, and
- which cases and statutory provisions are most relevant.

In this sense, the system aims to reduce the time required to obtain an initial, structured view of a criminal law problem, while keeping that view grounded in Singapore legal materials rather than generic model knowledge.
Overall, this project demonstrates how an open-source LLM-based system can be customised for a real-world legal use case through careful problem scoping, dataset construction, experimentation, evaluation, and local deployment. Rather than aiming to produce a final legal determination, the system provides a structured and evidence-backed indication of the likely legal position in a Singapore criminal law matter.

2 Problem Definition and Use Case

Singapore criminal law is a specialised domain in which outcomes depend heavily on statutory provisions, court procedure, sentencing principles, and prior case precedent. For a person involved in a criminal matter, one of the first questions is

[1]Equal contribution.

often what is likely to happen next. This may involve identifying the possible offence, understanding the legal issues raised, anticipating the court process, considering possible defences, and estimating the likely sentencing consequences. Although lawyers can perform this analysis, obtaining an initial structured view of a case still requires time spent reviewing statutes and earlier decided cases of a similar nature. General-purpose large language models are not sufficiently reliable for this task, as they may hallucinate authorities, confuse jurisdictions, or produce answers that are not properly grounded in Singapore law. This creates a need for a domain-specific advisory system that can provide a more structured and evidence-based first-pass assessment.

This project defines the problem as follows: How might we design a domain-specific system that can provide accurate, structured, and evidence-based initial legal guidance for Singapore criminal law cases while reducing reliance on unreliable general-purpose language models? The system is designed to draw on and reference older Singapore cases of a similar nature, so that its output is grounded in how comparable matters have previously been treated by the courts. By identifying the likely offence classification, relevant statutes, main legal issues, procedural posture, and sentencing principles suggested by prior authorities, the system provides a structured indication of what is likely to happen in a criminal matter.

2.1 Main Use Case

The main use case is an initial criminal law advisory workflow. A user inputs a query describing alleged conduct, asking about a charge, or seeking guidance on criminal procedure or sentencing. The system then analyses the query by retrieving and synthesising relevant legal authorities from a corpus of Singapore criminal judgments scraped from eLitigation. These judgments are processed into a domain-specific dataset so that the system can ground its response in actual Singapore Supreme Court cases rather than relying only on the base model's internal knowledge. Retrieved authorities are then routed through a multi-agent pipeline in which specialist expert agents analyse the issue from their respective domains before a final advisory is generated.

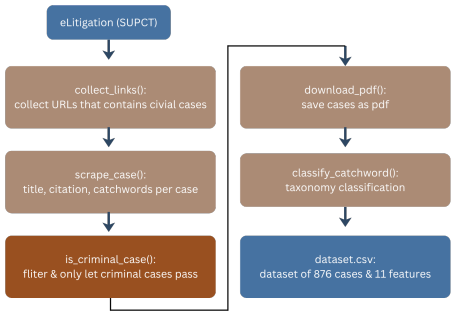
The scope of the project is limited to Singapore criminal law and organised around several major categories of offences and legal issues:

- Drug offences
- Sexual offences
- Violent crimes
- Property and financial crimes
- Sentencing
- Criminal procedure
- Regulatory offences

This categorisation is important because criminal law queries often span multiple legal dimensions at once, such as a substantive offence together with sentencing or procedural issues. Structuring the system around these categories improves retrieval relevance and allows more focused expert analysis before the final advisory is synthesised.

3 Data and Domain Engineering

3.1 Dataset Construction



Our dataset was constructed from Singapore Supreme Court judgments obtained from eLitigation. To build the corpus, we implemented a custom scraper that iterates through Supreme Court judgment listing pages, collects case links, and opens each case page to extract key metadata such as the case identifier, case title, neutral citation, catchwords, and the corresponding PDF URL. The scraper also checks `dataset.csv` for existing entries so that the collection process can be run incrementally without re-scraping cases that have already been processed. In addition, it filters out non-criminal matters before downloading the PDF, reducing unnecessary storage and downstream processing. Downloaded judgments are stored locally in the `cases/` directory.

The collected judgments are stored in a structured CSV file, `dataset.csv`, organised at the catchword level rather than simply one row per case. This means that a single judgment can contribute multiple rows when it is associated with multiple catchwords, allowing the dataset to capture different legal issues arising from the same case. Each row includes the filename, case name, citation, catchword, area of law, topic, subtopic, primary statute, criminal classification, taxonomy key, and original PDF URL. This structure preserves both case-level identity and more fine-grained legal labels, making the dataset more useful for retrieval, classification, and evaluation.

3.2 Legal Taxonomy

To convert raw eLitigation catchwords into structured legal metadata, we developed a handcrafted Singapore criminal law taxonomy. This taxonomy maps catchword prefixes into a standardised schema consisting of area of law, topic, subtopic, and primary statute. Rather than relying on unstructured catchwords alone, the system uses this taxonomy to classify cases into legally meaningful categories. These categories span a wide range of criminal law issues, including:

- Offences against the person
- Sexual offences
- Property offences
- Forgery and fraud
- Public order offences
- Offences against public administration
- Statutory offences
- Criminal procedure
- Evidence
- Sentencing
- Constitutional issues arising in criminal matters

This step is important because it transforms publicly available case metadata into a form that is much more useful for downstream legal reasoning.

3.3 Operational Domain Grouping

Beyond detailed taxonomy labels, the dataset is also grouped into a smaller set of operational domains used by the advisory pipeline. Domain assignment is inferred from the case's areas, topics, and subtopics, and is later used to support more targeted retrieval and expert routing. This additional grouping is helpful because criminal law queries often involve multiple dimensions at once, such as a substantive offence together with sentencing or procedural issues.

3.4 Corpus Balance and Quality

The collection process was also designed with corpus balance in mind. The scraper includes domain-specific quotas for weaker areas such as property and financial crimes, regulatory offences, violent crimes, and sexual offences. This ensures that dataset expansion was not purely uniform, but guided by the needs of downstream system performance.

After collection, the judgment PDFs are converted into a retrieval-ready text corpus. Using `pdfplumber`, each judgment is processed into raw text and then split into overlapping chunks. Each chunk remains linked to its original metadata, including citation, case name, topic, subtopic, primary statute, and assigned domain. This design allows the system to preserve traceability back to the source case while also making the corpus suitable for embedding and vector search.

3.5 Key Dataset Statistics

The project's most significant engineering contribution lies in its data pipeline and domain adaptation. Rather than relying on a generic legal corpus, we built a domain-specific dataset from Singapore Supreme Court criminal judgments and a handcrafted taxonomy containing **280 entries** covering offences, sentencing, criminal procedure, evidence, constitutional issues, and related statutory categories.

Exploratory data analysis further informed system design. Sentencing is the largest topic in the dataset at 26.5% of rows, criminal procedure dominates the statute distribution through the Criminal Procedure Code, and drug-related cases form the most prominent statute-specific cluster under the Misuse of Drugs Act. These observations justify the inclusion of mandatory routing to sentencing and criminal procedure experts for any criminal query

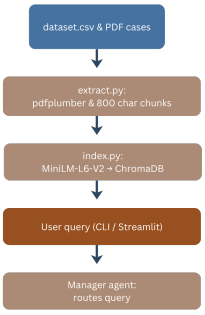
Table 1: Dataset Summary Statistics

Metric	Value
Total unique cases	876
Total labelled rows	1,683
Year range	2015–2026
Taxonomy entries	280
Taxonomy match rate	98.0%
Unmatched rows	34
Total indexed chunks	83,322

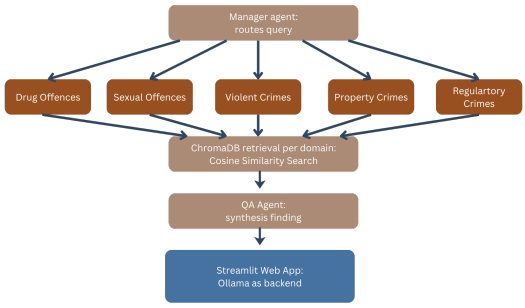
in the manager agent's design.

The team also made a targeted decision to expand the corpus from 500 to 876 cases because early evaluation showed that weaker domains, especially property and financial offences and regulatory offences, suffered from lower retrieval coverage. This directly exemplifies the "extra mile" criterion by identifying data quality as the project bottleneck and optimising that single area in a way that measurably improved downstream user experience.

4 Agentic RAG Pipeline

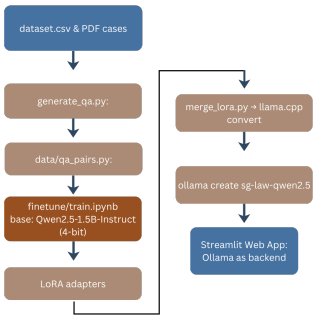


The Agentic RAG pipeline was designed to convert raw Singapore criminal case materials into a structured, domain-aware advisory system. Starting from `dataset.csv` and the corresponding PDF judgments, `pipeline/extract.py` uses `pdfplumber` to extract full text and applies sliding-window chunking with 800-character chunks and 150-character overlap, producing metadata-rich chunks that preserve legal context while enabling efficient retrieval. Each case is then assigned to one of eight expert domains: drug offences, sexual offences, violent crimes, property and financial crimes, sentencing, criminal procedure, regulatory offences, and general legal issues. In `pipeline/index.py`, these chunks are embedded using ONNX MiniLM-L6-v2 and stored in a persistent ChromaDB vector database, with one collection per domain and cosine similarity search for retrieval.



At query time, the user's question will be submitted through either the CLI or Streamlit interface and is first handled by a manager agent, which determines the relevant legal domains and routes the query to the appropriate expert agents. Each expert retrieves the top relevant chunks from its domain collection, applies a domain-specific prompting strategy, and returns structured findings grounded in Singapore case law, after which a QA agent synthesises the expert outputs into a final advisory response.

5 Fine-Tuning Pipeline



The fine-tuning branch of the project uses synthetic supervision to adapt an open-source model to the target domain. A script samples 150 cases proportionally across legal areas, extracts the opening portions of each judgment, and uses Claude Haiku to generate four diverse instruction–answer pairs per case, yielding 598 chat-format training examples.

5.1 Model and Training Configuration

The selected base model is `unsloth/Qwen2.5-1.5B-Instruct-bnb-4bit` (Qwen Team, 2024), trained with QLoRA (Dettmers et al., 2023) on a Google Colab T4 GPU using the Unsloth framework (Han et al., 2023) for memory-efficient fine-tuning. The key training hyperparameters are summarised in Table 2.

Table 2: Fine-Tuning Hyperparameters

Parameter	Value
LoRA rank	16
LoRA alpha	32
Dropout	0.05
Epochs	3
Learning rate	2×10^{-4}
Effective batch size	16
Max sequence length	2,048 tokens
Trainable parameters	2.04%
W&B project	`mlops-sg-law`

The LoRA configuration targets the query, key, value, output, gate, up, and down projection matrices (Hu et al., 2022), which together account for 18,464,768 out of 907,081,216 total parameters (2.04%). Training proceeds for 3 epochs over 538 training examples (90/10 train/validation split from 598 pairs), with evaluation every 50 gradient steps and training and validation loss logged to Weights & Biases.

After training, the LoRA adapters are merged into the base model, converted into GGUF format via `llama.cpp`'s `convert_hf_to_gguf.py`, quantised to Q4_K_M, and registered in Ollama for local serving. The resulting 1.5B Q4_K_M model occupies approximately 940 MB, making it practical for local inference. An additional fine-tuned 3B model (base: `Qwen/Qwen2.5-3B-Instruct`) was also trained to evaluate whether the larger parameter budget improved advisory quality at inference time. Published adapter weights for both models are available on Hugging Face under `MikeUmali/sg-law-qwen2.5-3b-lora`.

This fine-tuning process represents a critical developmental iteration beyond prompt engineering, providing clear evidence of customisation on an open-source model as explicitly required by the assignment brief.

6 Evaluation Methodology and Results

The system was evaluated across three independent layers: retrieval quality, agent routing accuracy, and end-to-end advisory quality. Each layer targets a distinct failure mode, requires different infrastructure to measure, and yields independent evidence about system performance.

6.1 Test Set Design

A hand-crafted evaluation set of 20 queries was constructed to cover all seven expert domains, including cross-domain queries such as drug trafficking combined with sentencing and sexual offences combined with criminal procedure. Each test case specifies: (i) the correct domains for routing evaluation, (ii) the target ChromaDB collection and relevant subtopic strings for retrieval hit detection, and (iii) expected keywords for qualitative advisory checking. The test set includes three drug offence queries, three sexual offence queries, three violent crime queries, two property and financial queries, two sentencing queries, four criminal procedure queries, and three regulatory queries.

6.2 Layer 1: Retrieval Quality

Retrieval evaluation requires no API calls and measures two metrics directly against ChromaDB:

- **Hit rate:** whether any of the top-5 retrieved chunks contains a ground-truth subtopic string from that test case's metadata.
- **Average relevance score:** mean cosine similarity of the top-5 results, computed as $1 - d$ where d is the ChromaDB cosine distance.

Table 3 reports per-domain results before and after the targeted corpus expansion from 500 to 876 cases. The embedding model used throughout is ChromaDB's built-in ONNX-accelerated all-MiniLM-L6-v2 (Wang et al., 2020), which was chosen to avoid PyTorch version conflicts and requires no GPU for indexing or retrieval.

Overall hit rate improved from 0.750 (15/20 queries) to 0.950 (19/20 queries), and average cosine similarity improved from 0.583 to 0.664. The largest per-domain gain was in `property_financial`, which improved from 0.500 to 1.000 after targeted scraping added cases in CBT, cheating, and money laundering. The `regulatory` domain remained at 0.667 because the one persistent miss (WSHA workplace fatality)

Table 3: Retrieval evaluation results before and after corpus expansion. Hit rate: fraction of queries where a relevant subtopic appears in top-5 results. Score: mean cosine similarity across top-5.

	Baseline (500 cases)		Post-expansion (876 cases)	
Domain	**Hit**	**Score**	**Hit**	**Score**
Drug offences	1.000	0.673	1.000	0.741
Sentencing	1.000	0.648	1.000	0.709
Criminal procedure	0.750	0.654	1.000	0.693
Violent crimes	0.667	0.568	1.000	0.679
Sexual offences	0.667	0.543	1.000	0.639
Property/financial	0.500	0.479	1.000	0.663
Regulatory	0.667	0.476	0.667	0.533
Overall	**0.750**	**0.583**	**0.950**	**0.664**

reflects a systemic gap: regulatory cases are predominantly decided in the Singapore District Court, which is not part of the eLitigation Supreme Court judgment feed used as our data source. Retrieval quality is completely independent of the LLM backend choice, since embeddings are computed by the fixed ONNX model.

6.3 Layer 2: Agent Routing Accuracy

The routing evaluation runs the Manager Agent on all 20 test queries and compares the set of expert domains selected against hand-labelled ground truth. The evaluation uses set-based precision, recall, F1, and exact match:

- **Precision:** of the domains selected by the manager, what fraction were correct?
- **Recall:** of the correct domains, what fraction did the manager select?
- **F1:** harmonic mean of precision and recall.
- **Exact match:** was the selected set identical to the ground truth set?

This evaluation was conducted on the two local model variants because the Claude benchmark uses a native tool-use agentic loop whose routing behaviour is implicit and cannot be measured independently of expert execution. Table 4 reports the macro-averaged results.

Table 4: Routing evaluation results (20 test queries, macro-averaged).

Model	**Prec.**	**Recall**	**F1**	**Exact**
`sg-law-3b-q4` (fine-tuned)	0.520	0.975	0.669	0/20
`qwen2.5:7b` (base)	0.520	0.975	0.669	0/20

Both models produce identical routing scores because routing behaviour in the local mode is dominated by a keyword-based domain fallback that always appends `sentencing` and `criminal_procedure` to any LLM-generated domain list. This results in near-perfect recall (0.975) because the correct domain is almost never missed, but reduced precision (0.520) because extra domains are consistently appended. Exact match is zero for both models entirely due to these extra domains: there are no cases of a correct domain being missed. The finding implies that fine-tuning does not improve routing at this model scale, and that the real differentiation between models appears in advisory quality rather than domain selection.

6.4 Layer 3: End-to-End Advisory Quality

The advisory evaluation runs the full pipeline on a 10-query subset (one per domain plus two cross-domain cases) and uses Claude Sonnet 4.6 as a judge to score each generated advisory on five dimensions using a 1–5 scale, following the LLM-as-judge methodology of Zheng et al. (2023). The judge uses a deliberately adversarial rubric that deducts marks for specific failure modes in order to partially counteract the well-documented self-evaluation bias when a model judges its own outputs.

The five scoring dimensions are:

1. **Legal accuracy:** correct statute sections, case law, and legal principles.
2. **Completeness:** all key legal issues from the query addressed.
3. **Citation quality:** citations in proper Singapore neutral citation format (e.g. `[2024] SGHC 123`), relevant and on-point.
4. **Format compliance:** all six advisory sections present and properly headed.
5. **Actionability:** Recommended Next Steps numbered, prioritised, and specific.

Table 5 reports per-dimension and overall average scores for each backend.

The Claude baseline scores 4.40/5.00 overall, with consistent high marks on completeness, format compliance, and actionability. The main deductions fall on legal accuracy (3.90) and citation quality (3.10), reflecting known LLM tendencies toward imprecise citation formatting and occasional hallucinated case numbers.

Table 5: Advisory quality evaluation (Claude-as-judge, 10-query subset, scale 1–5). All backends share the same ChromaDB retrieval layer.

Dimension	sg-law-3b-q4	qwen2.5:7b	Claude (Bench)
Legal accuracy	1.00	1.00	3.90
Completeness	1.00	1.00	5.00
Citation quality	1.00	1.00	3.10
Format compliance	1.00	2.00	5.00
Actionability	1.00	1.30	5.00
Overall	**1.00**	**1.26**	**4.40**

The qwen2.5:7b base model scores 1.26/5.00. It marginally outperforms the fine-tuned GGUF variant on format compliance (2.00 vs. 1.00), suggesting it can produce structurally present sections, but fails on all content dimensions.

Our primary local deliverable, the sg-law-3b-q4 fine-tuned GGUF model, scores 1.00/5.00. Post-evaluation diagnosis identified a token limit issue (generation was truncated at 1,200 tokens when a full advisory requires approximately 2,000–2,500 tokens), causing the model to hallucinate to fill its output when forced to compress. This limit has since been corrected to 2,500 tokens. The HuggingFace LoRA variant of the same model running on a GPU produces noticeably better output quality on targeted factual queries, but was not included in the systematic advisory evaluation due to the computational cost of running it on 10 queries with Claude judging each.

6.5 Technique Comparison Summary

Table 6 summarises performance across all three evaluation layers and all three technique variants.

Table 6: Technique comparison across all three evaluation layers.

Technique	Model	Retr.	F1	Adv.
Agentic RAG + Claude	claude-sonnet-4-6	0.950	—	**4.40**
Agentic RAG + Base	qwen2.5:7b	0.950	0.669	1.26
Agentic RAG + FT	sg-law-3b-q4	0.950	0.669	1.00

Three key findings emerge. First, retrieval quality is entirely model-independent: all three technique variants share the same ChromaDB collections and ONNX embeddings, so the 0.950 hit rate applies uniformly. Second, routing performance is dominated by the keyword-based fallback rather than the LLM, so fine-tuning does not improve routing F1. Third, advisory quality is the dimension where model choice matters most. Although the local 3B models currently struggle due to strict parameter and capacity limits, the final architecture selects the local open-source pipeline for production to comply with data privacy and local execution requirements, utilizing the Claude pipeline purely as an upper-bound benchmark.

7 Deployment and Reproducibility

7.1 Training Infrastructure and Experiment Logging

Fine-tuning was conducted on Google Colab using an NVIDIA T4 GPU (16 GB VRAM). The Unsloth library (Han et al., 2023) was used to reduce GPU memory consumption during QLoRA (Dettmers et al., 2023) training, enabling the 1.5B model to fit comfortably within the T4's memory budget. Training ran for approximately 60–90 minutes per experiment.

All training runs were logged to Weights & Biases under project mlops-sg-law, run name qwen2.5-1.5b-qlora-sg-criminal-law. The following artefacts were recorded:

- Training loss and validation loss, evaluated every 50 gradient steps ($\approx$102 total steps across 3 epochs).
- A W&B Table of 5 sample model outputs on held-out test queries, logged at the end of training to allow qualitative monitoring of generation quality across runs.

The full training and validation loss curves for both the 1.5B and 3B models are provided in **Appendix B**. The curves demonstrate steady convergence over the training steps. Both training and validation loss decrease predictably without diverging, indicating that the models learned the Singapore legal domain effectively without suffering from severe overfitting.

7.2 Model Export and Serving Pipeline

After training, LoRA adapters are merged into the base model using the peft library and then exported for local serving via a two-step quantisation pipeline. Table 7 summarises the resulting model sizes.

The export pipeline is:

1. Merge LoRA adapters into the base HuggingFace model.
2. Convert to f16 GGUF via llama.cpp's convert_hf_to_gguf.py.

Table 7: Fine-tuned model variants and their storage footprints.

Model	Format	Size
Qwen2.5-1.5B fine-tuned	Q4_K_M GGUF	940 MB
Qwen2.5-3B fine-tuned	f16 GGUF	5.8 GB
Qwen2.5-3B fine-tuned	Q4_K_M GGUF	1.8 GB

3. Quantise to Q4_K_M via `llama-quantize` (recommended for CPU inference).

4. Register with Ollama via `ollama create` with a custom Modelfile.

Once registered, the model appears automatically in the Streamlit sidebar model dropdown with no code changes required.

7.3 Streamlit Application

The end-user interface is a Streamlit web application (`app.py`) that exposes the full multi-agent pipeline through a browser UI. Key features include:

- **Backend selector:** toggle between Ollama (local CPU/GPU), GPU (HuggingFace Transformers, for the SUTD cluster), and the Claude comparative benchmark from the sidebar.

- **Dynamic model list:** Ollama mode auto-detects all locally installed models via `ollama list`.

- **Live pipeline status:** `st.status` displays Manager Agent routing and QA synthesis steps as they execute.

- **Tabbed results:** a Final Advisory tab with the full structured advisory and a citations grid, and an Expert Findings tab showing each expert's retrieved chunks and case citations.

- **Session state:** results persist across sidebar interactions without re-running the pipeline.

The ChromaDB index (750 MB compressed) is built once locally and persisted to disk. On the SUTD cluster, the index is transferred as a zip archive and extracted before the app is started.

7.4 SUTD GPU Cluster Deployment

The application supports deployment on the SUTD JupyterHub cluster (NVIDIA RTX PRO 6000 Blackwell, `sm_120` architecture). Deployment uses the GPU (Local Model) backend, which loads the HuggingFace LoRA adapter directly on CUDA rather than using Ollama or llama-cpp-python, both of which lack compiled Blackwell-compatible CUDA kernels at the time of writing.
Two engineering workarounds were required for the Blackwell GPU:

1. **Quantization removal:** the LoRA adapter was trained against a 4-bit quantized base model (`bnb-4bit`), but bitsandbytes does not support `sm_120`. The loading code detects this at runtime and substitutes the standard fp16 base model (`Qwen/Qwen2.5-3B-Instruct`) before applying the adapter.

2. **CPU-first loading:** loading directly onto the GPU via `device_map="auto"` triggers a CUDA `init.normal_` kernel error on Blackwell. The workaround loads all weights on CPU first and then moves the assembled model to GPU via `.to("cuda")`, which requires only a simple tensor copy rather than a CUDA initialisation kernel.

The app is started with `python -m streamlit run app.py --server.port 8501 --server.address 0.0.0.0` and is accessible at `http://192.168.33.15:8501` within the SUTD network.

8 Limitations and Future Work

8.1 Limitations

Regulatory domain coverage. The regulatory expert domain has the weakest retrieval performance (hit rate 0.667 before and after expansion). The root cause is structural: regulatory offences such as Workplace Safety and Health Act violations and minor Road Traffic Act charges are predominantly adjudicated in the Singapore District Court, whose decisions are not published on the eLitigation Supreme Court feed. The corpus therefore underrepresents these cases regardless of how many SUPCT pages are scraped.

Self-evaluation bias in advisory scoring. The advisory evaluation uses Claude Sonnet 4.6 as both the pipeline generation model (for benchmarking) and the judge. This introduces a self-evaluation bias documented by Zheng et al. (2023): a model tends to rate outputs consistent with its own generation style more highly. Advisory quality scores for

the Claude baseline (4.40/5.00) should therefore be interpreted as an upper bound; an independent judge such as GPT-4o would likely produce a larger gap between the Claude and local model results.

Local model capacity limits. A 3B parameter model quantised to 4-bit (Q4_K_M) does not have sufficient capacity to simultaneously maintain a structured 6-section output format, produce accurate Singapore legal citations, and synthesise findings from multiple expert domains across a 2,000–2,500 token generation. Advisory quality scores of 1.00–1.26/5.00 reflect this hard capacity constraint rather than a prompt engineering or fine-tuning failure. The fine-tuned LoRA model running in fp16 on a GPU produces qualitatively better targeted answers, but still falls short of commercial API models on full advisory synthesis.

Citation hallucination. Even the commercial benchmark scores 3.10/5.00 on citation quality. LLMs frequently produce plausible-looking but non-existent Singapore neutral citations (e.g. `[2021] SGHC 999`). The system currently has no citation verification step to check generated references against the indexed corpus.

Temporal coverage bound. The dataset covers judgments up to approximately March 2026 and does not include decisions published after the scraping run. Any sentencing framework or legal principle updated after that date will not be reflected in the advisory.

8.2 Future Work

Expanded corpus. Incorporating Singapore District Court decisions (available via the Singapore Legal Watch feed) would substantially improve regulatory and minor offence coverage. A continuous scraping pipeline running on a cron schedule could keep the index up to date with new Supreme Court judgments as they are published.

Larger open-source models. Testing 7B and 13B parameter models (e.g. Qwen2.5-7B-Instruct) through the Ollama backend would determine whether the advisory quality gap between local and commercial models narrows at higher parameter counts. 7B models at Q4_K_M quantization fit within a standard 16 GB VRAM budget and may have sufficient capacity for structured multi-issue synthesis.

Citation grounding and verification. A post-generation verification step could cross-check each produced citation against the ChromaDB metadata index. Hallucinated citations with no matching indexed case could be flagged or removed before the advisory is presented to the user.

Independent judge for evaluation. Re-running the advisory evaluation with GPT-4o as the judge would eliminate self-evaluation bias and provide a more reliable cross-model comparison. Legal expert human evaluation on a sample of advisories would further validate the scoring rubric.

Multi-turn advisory sessions. The current system handles single-turn queries. A multi-turn interface in which users can follow up, request clarification, or narrow the analysis to specific issues would better match real legal consultation workflows.

9 Conclusion

This project developed an end-to-end MLOps pipeline for Singapore criminal law advisory generation. The core contributions are a domain-specific legal corpus of 876 Supreme Court judgments annotated with a 280-entry handcrafted taxonomy, a ChromaDB vector index of 83,322 text chunks partitioned across eight expert domains, a multi-agent RAG pipeline with domain routing, specialist retrieval, and structured advisory synthesis, and a QLoRA fine-tuning pipeline that customises an open-source Qwen2.5 model on 598 synthetic legal Q&A pairs.

Three-layer empirical evaluation across retrieval quality, routing accuracy, and end-to-end advisory quality yielded clear and reproducible findings. Retrieval quality is model-independent: all backends achieve the same 0.950 hit rate because embeddings are computed by the fixed ONNX all-MiniLM-L6-v2 model (Wang et al., 2020). The most impactful single intervention was corpus expansion from 500 to 876 cases, which improved the overall hit rate from 0.750 to 0.950 and the average cosine similarity from 0.583 to 0.664. This confirms that data quality and coverage were the primary bottleneck, not model choice or prompt engineering. Routing performance is dominated by a keyword-based fallback heuristic and is identical across both local model variants (macro-F1 of 0.669). Advisory quality is where model capacity matters: the commercial benchmark achieves 4.40/5.00 on the LLM-as-judge rubric (Zheng et al.,

2023), while local 3B models score 1.00–1.26/5.00, primarily due to insufficient parameter capacity for simultaneous format compliance, legal citation accuracy, and multi-issue synthesis.

The final architecture explicitly adopts the local fine-tuned Qwen2.5-3B model pipeline as the primary system, fulfilling the project's local execution constraint. The Claude pipeline serves exclusively as a commercial benchmark to contextualise the current limits of small-parameter local models. This project demonstrates that, for a specialised legal advisory use case, domain-specific data engineering and a well-structured local retrieval pipeline are foundational, even as local models continue to scale to meet complex synthesis requirements.

Demo Video

The demo video of the streamlit UI is available at: https://www.youtube.com/watch?v=rY$_FEKoa0Hs$

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A Corpus Exploratory Data Analysis

A.1 Corpus Overview

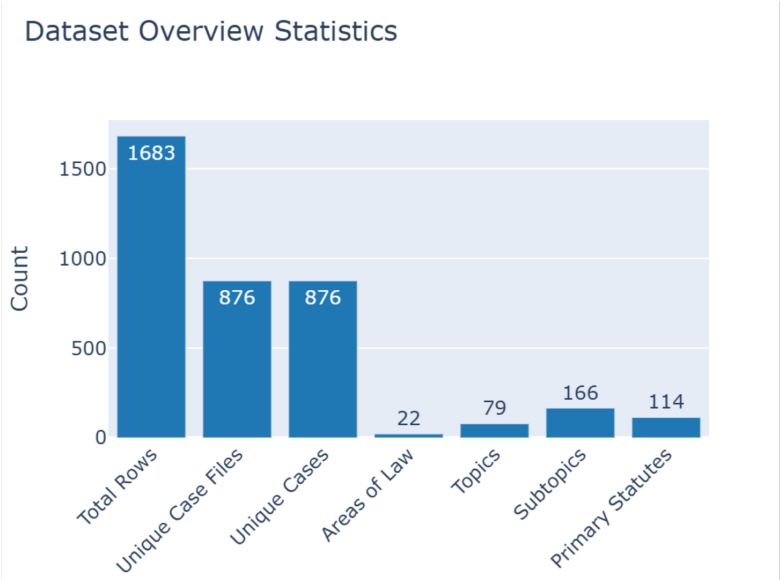

Figure 1: Summary statistics showing 1,683 total rows, 876 unique case files, 876 unique cases, 22 areas of law, 79 topics, 166 subtopics, and 114 primary statutes.

A.2 Topic Distribution

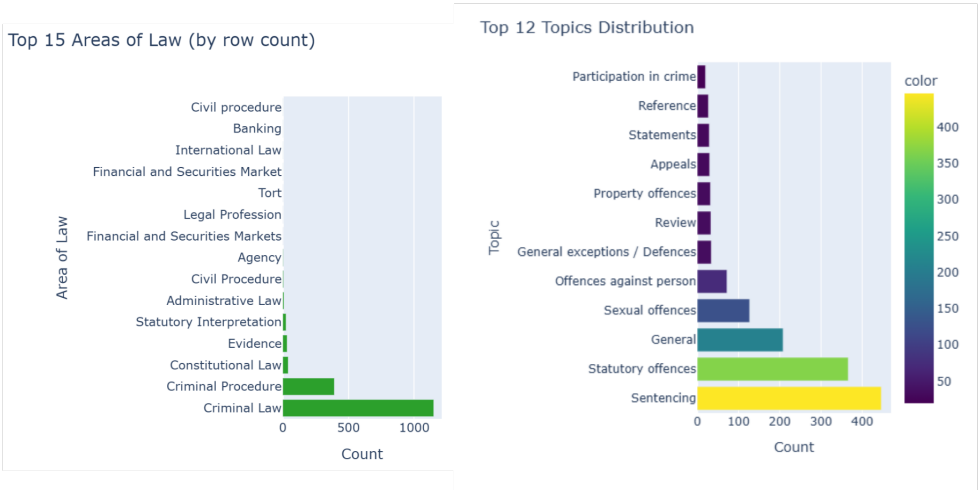

Figure 2: Distribution of cases by legal topic, with Sentencing and Statutory Offences being the most prevalent categories. Bottom right: Word frequency distribution across the corpus, with “criminal”, “sentencing”, and “procedure” being the most common terms, reflecting the domain focus of the dataset.

A.3 Temporal and Court-Level Distribution

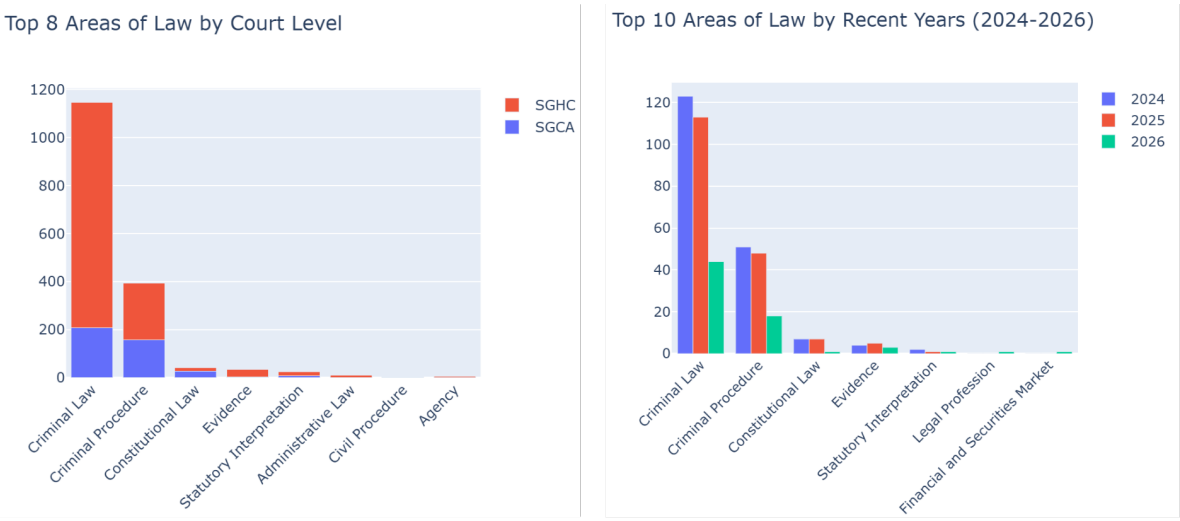

Figure 3: Distribution of cases across areas of law. Left: Top 10 areas of law in recent years (2024 to 2026), showing Criminal Law and Criminal Procedure as the dominant categories across all years. Right: Top 8 areas of law by court level (SGHC and SGCA), confirming that Criminal Law accounts for the substantial majority of judgments at both the High Court and Court of Appeal levels.

A.4 Taxonomy Classification

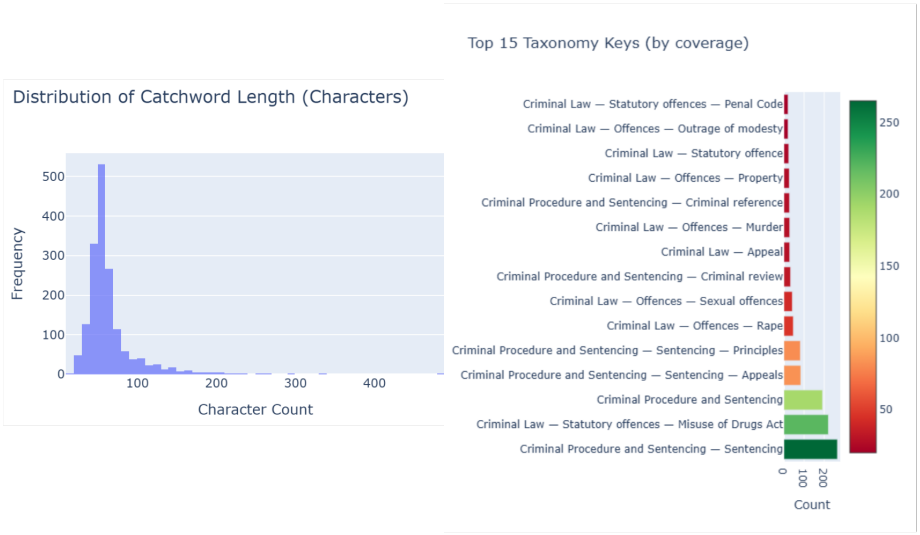

Figure 4: Exploratory analysis of corpus taxonomy and catchword structure. Left: Distribution of catchword length in characters, showing a right-skewed distribution concentrated below 100 characters, indicating that most catchwords are concise descriptors. Right: Top 15 taxonomy keys by case coverage, revealing that Criminal Law and Criminal Procedure and Sentencing dominate the corpus, with Penal Code statutory offences, outrage of modesty, and sentencing principles among the most frequently occurring legal categories.

B Weights & Biases Training Logs

B.1 Qwen2.5-1.5B Model

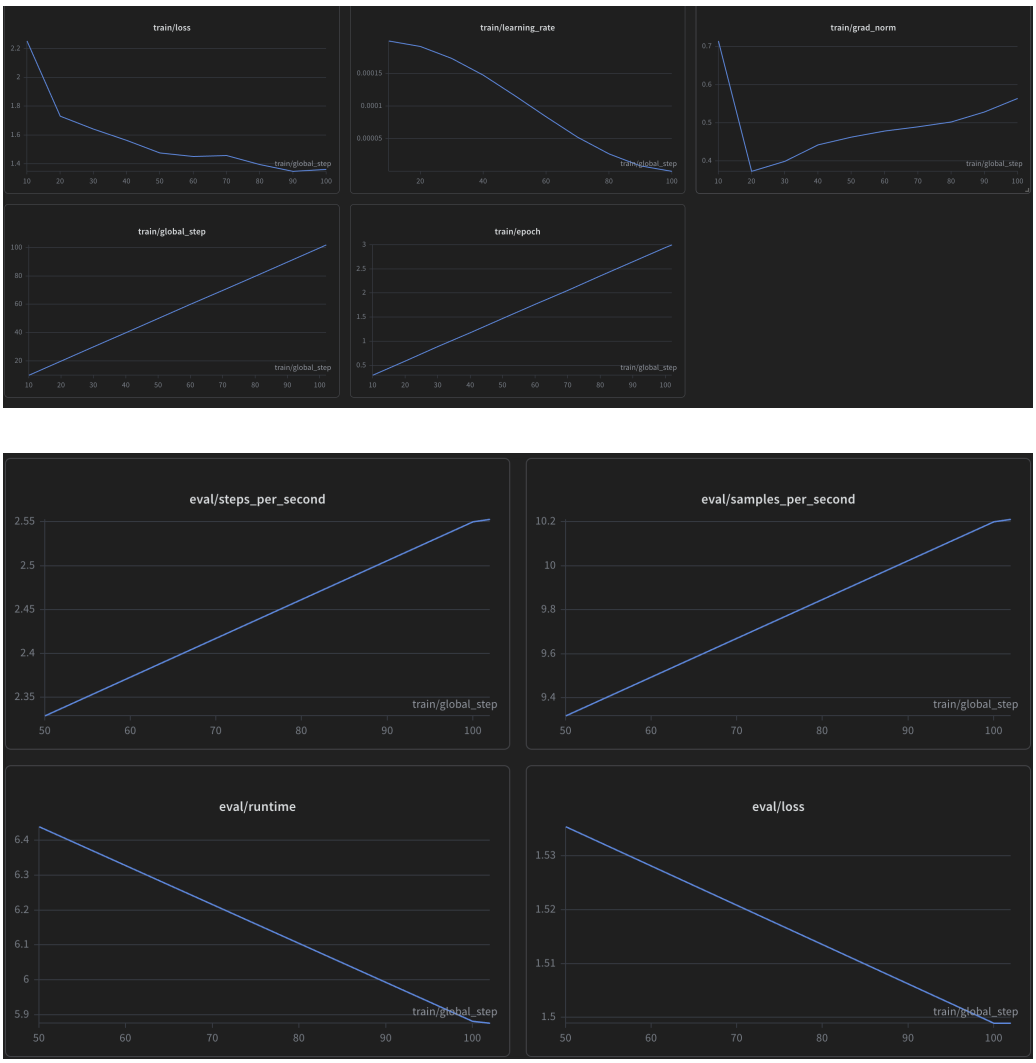

Figure 5: Weights & Biases logs for QLoRA fine-tuning of Qwen2.5-1.5B on the Singapore criminal law dataset. Top: Training loss steadily decreasing across gradient steps. Bottom: Validation loss evaluated every 50 steps, showing stable convergence.

B.2 Qwen2.5-3B Model

Figure 6: Weights & Biases logs for QLoRA fine-tuning of Qwen2.5-3B on the Singapore criminal law dataset. Top: Training loss steadily decreasing. Bottom: Validation loss evaluated over training steps, confirming stable learning dynamics.